

independent contribution suggested that testing of assumptions as such was not an important issue. However, such a proposition did lead to much controversy as we shall see in the next section.

4.5 MILTON FRIEDMAN AND INSTRUMENTALISM IN ECONOMICS

Milton Friedman's "The Methodology of Positive Economics", the lead article in his book *Essays in Positive Economics* (1953), is among the best-known pieces of methodological writings in economics. For over two decades the methodological prescriptions advanced in his essay became widely accepted among many working economists, in spite of wide controversy it generated. Rightly, as observed by Caldwell, it was a remarkable masterpiece of marketing certain notions of explanation as the basis of the methodology of positive economics. In what follows, we shall discuss in brief Friedman's contribution including some of the major points of criticism. Friedman sets out by stating that the purpose of his essay is to counter criticism of two pillars of neo-classical, economics viz., 1) the maximization behaviour and 2) The model of perfect competition. He goes on to establish the methodological foundations of neo-classical economics by posting his own version of positivism which later came to be branded as "instrumentalism".

4.5.1 Goal of Science

According to Milton Friedman, the ultimate goal of positive science "is the development of a 'theory' or 'hypothesis' that yields valid and meaningfulpredictions about the phenomenon". The criteria for acceptability of theory or hypothesis are three:

- 1) that theory / hypothesis be logically consistent and contain categories which have meaningful empirical counterparts;
- 2) that there are 'substantive hypotheses' which are testable; and
- 3) that "the only relevant test of the validity of a hypothesis is comparison of its predictions with experience".

Further, when an infinite number of hypotheses generally consistent with an observed set of facts generally exist, other criteria-simplicity and fruitfulness-be invoked to choose among competing hypotheses.

4.5.2 Relationship Between Significance of Theory and 'Realism' of Assumptions

Friedman is critical of those who instead of testing a theory for predictability, try to evaluate theory by the 'realism of their assumptions'. For Friedman, disparaging a theory for having 'unrealistic assumptions' is ridiculous. He tries to show that most significant theories are actually characterized by descriptively inaccurate assumptions. He observes: "Truly important and significant hypotheses will be found to have "assumptions" that are widely inaccurate, descriptive representations of reality, and, *in general, the more significant the theory, the more unrealistic the assumptions*". (emphasis added).

Realism of assumptions does not matter except when certain hypotheses are derived from assumptions, and when there is no test available, realist description is in order. With these two exceptions, realism of assumptions do not matter for the following reasons:

- 1) Theory is supposed to lead to simplification of complex reality by abstraction. There is inverse relationship between realism and abstraction. The more realism one insists upon, the more abstract and the complex the explanation.
- 2) The uninformativeness of knowledge of discrepancies between assumptions and facts, makes one to treat realism of assumptions as of no importance. He cites the example of Galileo's Law of Falling Bodies:

"If a body falls toward the earth in vacuum, its instantaneous acceleration is constant, $s = \frac{1}{2}gt^2$. The realism of vacuum need not be tested. Since the prediction is true, then the assumption be treated as if true. Turning to economic theory, Friedman cites the "Billiard Ball Player" example and where once the ball reaches the pouch one could assume as if the player knew all the laws of time and motion and extends it to profit maximization: 'under behave as if they were seeking rationally to maximize their expected returns.....and have full knowledge of the data needed to succeed in this attempt'".

- 3) The presence of 'undesigned' classes of implications can be avoided by *as if* assumption.

Milton Friedman concludes his essay by asserting unity of method in all positive sciences. For him there are no methodological differences between natural and social sciences. He regards economics as "objective" as physical sciences. His main conclusions are:

- 1) that realism of assumptions is "largely" irrelevant to validation of theories, which ought to be judged 'almost' solely in terms of their instrumental value in generating accurate predictions.
- 2) standard economic theory has an excellent predictive record as judged by countless application to specific problems.
- 3) the dynamics of competition over time accounts for this splendid track record.

4.5.3 Critical Appraisal of Friedman's Instrumentalism

Milton Friedman's methodological approach is criticized widely. The major criticism relates to the proposition that realism of assumptions does not matter. Friedman's assertion that not only the realism of assumptions is necessary but "in general, the more significant the theory, the more unrealistic the assumptions" is called by Samuelson as a flamboyant exaggeration, as the extreme version of "F-Twist" (meaning Friedman-Twist). It is one thing to say that proving realism of all assumptions is difficult, but altogether different to say that further the assumptions move away from realism, closer is the theory to predictions. Even Fritz Machlup, L. Qsoland and Nagel, who were generally in defence of Friedman's methodology, would insist on the realism of at least second order assumptions. It is pointed out that direct evidence about assumptions is not

necessarily more difficult than testing a theory. Test assumptions may yield important insights into the theory. Caldwell points out that philosophically, Friedman's insistence on prediction not explanation may result in correlation not causation, i.e., it may result in "measurement without theory". Blaug points out that accurate predictions are not the only relevant test of the validity of a theory and it would be impossible to distinguish between genuine and spurious correlations. Further, Friedman's instrumental attitude to theories, ignores 'truth value' of theories as many philosophers of science would insist upon.

Check Your Progress 2

- 1) Identify the important features of methodological view point of L. Rabins.

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- 2) What is the important contribution of Hutchison in the area of logical empiricism?

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- 3) Which criteria have been laid down by Milton Friedman for acceptability of theory or hypothesis?

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- 4) Identify the basis of the major criticism against Friedman's instrumentalism proposition.

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4.6 PAUL SAMUELSON AND OPERATIONALISM

Paul Samuelson's methodological contribution, though not comparable to his substantial work in economic theory, does offer certain insights into the turn of economics towards logical empiricism. Samuelson is against a **priorism**. He is more akin to logical empiricism of Hutchison's kind. He does show the influence of Popper's variety of 'rational reconstruction'. His contribution to economic methodology could be seen in terms of two theses, which he advances. One has come to be described as 'operationalism' and the other as 'descriptivism'. We shall elaborate these two theses.

4.6.1 Operationalism

According to him, economists should seek to discover ‘operationally meaningful theorems’. Theories are strategically simplified description of observable and refutable empirical regularities. He begins with the criticism of Friedman for the latter’s F-Twist on realism of assumptions and emphasizes the need for methodological clarity. He proposes the thesis of “logical equivalence” as the basis for methodological consistency. According to “logical equivalence”, theories are merely equivalent restatements of assumptions and conclusions, i.e., $A=B=C$, where A is defined as ‘assumptions’, B is defined as theory and C as consequences or predictions.

Theory (B) consists of “a set of axioms, postulates or hypotheses that stipulate something about observable reality...” “The set is either confirmable or refutable in principle by observation. A theory has a set of consequences (C) which are logically implied by theory, and a set assumptions (A) which logically implies the theory. The degree of “realism”, “factual correctness”, “empirical validity” or “truth” of any one of A,B,C is shared by the other two. Referring to ‘F-Twist’, he observes first, it is a contradiction to maintain all consequences (C) are valid and the theory (B) and the assumptions (A) are not valid. Second, it is absurd to maintain, in case where only some of the consequences (C) are valid, that the theory (B) and assumptions (A) are important though invalid. The part of the theory set and assumptions set corresponding to the invalid part of the consequence set should be eliminated.

4.6.2 Descriptivism

Samuelson has certain exalted view of explanation, which he considers different from the usual notion in science. He observed, “scientists never” explain” any behaviour, by theory or any other hook. Every description that is superseded by a “deeper explanation” turns out ...to have been replaced by still another description” An explanation, as used legitimately in science, is a better kind of description and not something that goes ultimately beyond description. It is this emphasis and elaboration of ‘description’ that earns Samuelson’s approach the title of ‘descriptivism’. Why only description? First, a theory is just description of observable experience, a convenient mnemonic representation of empirical reality. Second, knowledge consists essentially of observational reports. A theory expressible in observational language is superior to those which are not. Explanations are ultimate. Apriorisms must be avoided; hence theories should be expressed in observational language. All known theories in science are expressible in terms of observational statements, i.e., basic statements.

Samuelson is criticized for not practicing what he preaches methodologically. Machlup, citing Samuelson’s famous work on ‘factor price equalisation’, accuses him for not following his own norm of deriving “operationally meaningful theorem” based on unrealistic assumptions. Another famous criticism is that of Stanley Wong’s against the methodological foundations of Samuelson’s revealed preference theory.

4.7 THEORY – ASSUMPTIONS DEBATE IN ECONOMICS: A LONG VIEW

The spectacular advance of modern science is usually attributed very largely to the development of deductively related statements known as theories. Theories in science brought improvements in order and clarity, broadened the scope of generalizations and scored extraordinary predictive success. However, economic theory does not command as much respect as theories in physical sciences. Ever since the classical economics, there has been criticism of unrealistic assumptions of economic theory. There have been reasoned replies from time to time. Jack Melitz (1965), in one of his classic papers on the theory – assumptions controversy, provides an account of the response to the criticism since late Twentieth century. Here is a brief summary, which helps us to understand the problems of explanation that persist in economics.

During 1880-1920 advocates of economic theory adopted a *moderate and conciliatory* stance. They agreed that economic theory made false assumptions, and admitted that the value of the theory depended greatly on the degree of accord between the assumptions and the facts. Yet they insisted, first, that the assumptions did correspond broadly with events. Second sacrifice of some accuracy for simplicity was justified in view of complexity of reality. Further, they emphasized importance of combining the use of simplifying assumptions with protective measures:

- 1) The need to pursue more inductive studies in all areas of economics.
- 2) The determination of reasonable proximity between major assumptions and facts before application of economic theory.
- 3) The alteration of assumptions to suit the particular case involved.

During 1920s and 1930s with notable advance of purely logical branch of economics, the advocates of economic theory lost appeal. The practitioners of economic theory with the support of Robbins and the Austrians turned to a *priorism*. There was a stance that economic theory is its own reward.

In 1948-1953, as we have seen earlier, Milton Friedman tried to supply a logical foundation for the developing attitude that the realism of assumptions is not a genuine, or only a secondary concern. False assumptions in economics do not result in a handicap. Assumptions must simply work, and yield reliable results. The criticism that assumptions in economics do not correspond with facts is besides the point. In 1955 Fritz Machlup joined forces with Friedman, claiming to bring with him the support of experts in the philosophy of science and the whole *tradition* of political economy. The result is that economics continues to live with a certain methodological inadequacy. The other reasons for this disquiet in explanation in economics are summed up in the next section.

4.8 AMARTYA SEN ON HETEROGENEITY OF EXPLANATION IN ECONOMICS

By now it is clear that explanation in economics is nowhere near being in a satisfactory state. In a broad based critique of contemporary economic

methodology, Amartya Sen (1989) draws attention to the heterogeneity of the subject matter of economics. Any attempt to think of mono-method for all the diverse concerns of economics is bound to cause the kind of disquiet that we experience today. He cogently argues for heterogeneity of methodological approaches in economics.

4.8.1 Heterogeneity of Substance and Methods of Economics

According to Sen, economics as a subject is concerned with many different types of problems. The diversity of the discipline of economics should be kept in view to achieve an adequate grip on the methodological issues in the subject. The subject of economics includes three diverse, though interrelated, exercises:

- 1) Predicting the future and causally explaining the past events.
- 2) Choosing appropriate descriptions of states and events in the past and the present, and
- 3) Providing normative evaluations of states, institutions, and policies.

These exercises are interrelated but each requires a different methodological approach. For instance, the method of scientific explanation that insists on prediction is concerned only with the first set of exercises. The 'methodology of economics' has to admit enough diversity to be able to deal with other classes of problems as well.

4.8.2 Testing and Verification

Testing and *Verification* are important for many types of economic analyses since they are concerned with causal relationships and with making predictions. But these are not suitable for all economic theories. For instance evaluative exercises are not open to testing. Normative evaluation is a different discipline from that of making predictions on the basis of causal hypotheses. Similarly, some descriptive propositions do not have predictive content, and "testing" would be the wrong operation to seek. As far as causal theories are concerned, the need for testing them with empirical information is fairly accepted in principle by economists. Conceptual and analytical issues need to be explored very substantially to understand what types of relationships might be involved. Analyses at this stage are not meant for testing and verification. Of course, one should not end at this stage and move up to the next where testing is possible.

4.8.3 Value Judgments and Welfare Economics

Sen draws attention to the fact that following Robbins value judgments are kept out of economics. "The decision to keep economics 'value-free' would, of course, militate against the subject of welfare economics as such". Welfare economics still is accepted as important and in this domain to keep economics value-free may not be a value that would be appreciated. In evaluative exercises in welfare economics descriptive methods becomes indispensable.

4.8.4 Formalisation and Mathematics

Mathematics helps to an extent in formal reasoning and helped in systematization of many economic propositions. However, there are severe limitations of formal language of mathematics. Not all economic propositions can be reduced into mathematics. Lack of balance often has resulted in certain degree of over-

formalization of economics. While recognizing the positive contribution of mathematics in lending rigour to certain economic propositions, the negative contribution is through over concentration on mathematics to the exclusion of other disciplines. Such excesses of formalization need to be corrected.

Check Your Progress 3

- 1) What do you mean by the term 'logical equivalence'?

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- 2) Do you agree that different methodological approaches are needed in Economics?

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- 3) To what extent mathematics should be allowed to use in systematization of economic proposition?

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4.9 LET US SUM UP

Keeping in view that the subject matter of economics is complex and heterogeneous, any single model of explanation is bound to face serious limitations. Yet there has been an ambition in the profession from the times of classical political economy, which has turned towards a kind of arrogance that economics is a science like any physical sciences and should conform to the rule of one methodological approach, viz., logical positivism. This has been subjected to severe criticism on its limitations.

The mainstream economics is not able to come out of this obsession "It is about the one methodological rule which has dominated economics since early 1960s. The rule at issue is the methodological requirement that economic models or theories, if they are going to be given serious consideration by practicing economists, must be shown to be testable, where successful test of theory is defined as a falsification of that theory. A testable theory is a falsifiable theory".

The exploratory lesson on explanation that we had should help a student of economics to move out of this obsession. It is not that abandon testing but we must abandon the habit of considering evaluation, description, and even advocacy as not being part of pursuing knowledge in economics. Heterogeneity of economics needs diversity of approaches if the disquiet of disaffection that pervades economics is to be overcome.

4.10 KEY WORDS

- Determinism** : A term used to describe an argument or methodology that simply reduces causality to a single set of factors acting more or less directly to produce outcomes.
- Hypothesis** : Hypothesis refers to a tentative statement that can be tested by applying the methods of particular science.
- Empiricism** : Empiricism is generally regarded as being at the heart of the modern scientific method that our theories should be based on our observations of the world rather than on intuition or faith; that is, empirical research and a posteriori inductive reasoning rather than purely deductive logic.
- Ideology** : A systematic body of concepts about human life or culture; a manner or the content of thinking characteristic of an individual or group or culture; also refers to the integrated assertion and theories that constitute a socio-political programme.
- Methodological Dualism** : Referring to positivist belief in separation of the knower from the known or of the subject from object.
- Methodological Monism** : In contrast to the compartmentalization of dualism, monism views the world as a “seamless web”. In terms of Gouldner’s argument, the separation between the knower and the known must be overcome.
- Nomothetic** : Relating to discovery of general laws or relating to the discovery of universal law.
- Pseudo Science** : Pseudo science refers to any body of knowledge or practice, which purports to be scientific or supported by science but which is judged to fall outside the domain of science.

4.11 SOME USEFUL BOOKS

Blaug, Mark (1980); *Methodology of Economics*, Cambridge University Press, Cambridge.

Caldwell, Bruce (1982); *Beyond Positivism: Economic Methodology in the Twentieth Century*, George Allen and Unwin, London.

Friedman, Milton (1953); *Essays in Positive Economics*, University of Chicago Press, Chicago.

Melitz, Jack (1965); “Friedman and Machlup on the Significance of Testing Economic Assumptions”, *Journal of Political Economy*, pp.37-60.

Sen, Amartya (1989); “*Economic Methodology: Heterogeneity and Relevance*”, *Social Research*, Vol.56, No.2 (Summer 1989), pp. 299-329.

4.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Positive, abstract and deductive briefly called hyupothetico-deductive model of explanation.
- 2) See Sub-section 4.2.2.
- 3) Methodological pluralism refers to comprehensive reconciliation of different methods and acceptance of the idea that different methods are appropriate depending upon the object in view, the stage of investigation reached and the material available.

Check Your Progress 2

- 1) Stress on pure theory bases on a priori truth of experience and without any need for testing, no room for ethical consideration, rationality as maximizing behaviour and as consistency of choice.
- 2) The insistence of testing both theory as well as assumptions.
- 3) See Sub-section 4.5.1
- 4) The proposition that realism of assumption does not matter.

Check Your Progress 3

- 1) Theories are merely equivalent restatements of assumptions and conclusions.
- 2) See Section 4.8
- 3) See Sub-section 4.8.4

UNIT 5 FOUNDATIONS OF QUALITATIVE RESEARCH: INTERPRETATIVISM AND CRITICAL THEORY PARADIGM

Structure

- 5.0 Objectives
- 5.1 Introduction
- 5.2 Interpretive Paradigm
- 5.3 Critical Theory Paradigm
- 5.4 Applications in Research: Illustrative Cases
- 5.5 Let Us Sum Up
- 5.6 Exercises
- 5.7 Key Words
- 5.8 Some Useful Books
- 5.9 Answers or Hints to Check Your Progress Exercises

5.0 OBJECTIVES

This unit introduces the two widely used and well-established paradigms namely, interpretive and critical theory. Unlike post-positivism, these paradigms are the pillars of qualitative research. After going through this unit, you will be able to:

- describe the essence of interpretive and critical theory paradigm;
- discuss their theoretical framework;
- elucidate the nature of reality and methodology underlying these two paradigms; and
- state the examples and applications of interpretive and critical theory paradigm in economics.

5.1 INTRODUCTION

As has been discussed earlier in Unit 1, you have come to know that a paradigm essentially refers to a comprehensive belief system or theoretical framework/world view that guides research and practice in a particular field. At its basic level, it has a philosophy of science that makes a number of fundamental assumptions about the nature of truth and what it means to know. Unlike positivism, that is based on empiricism and deductive reasoning for verification of truth, interpretive and critical theory paradigms are based on **subjectivism** and **substantialism** for the search of truth. The underlying ontological and epistemological assumptions of interpretivism and critical theory paradigms have been discussed in the subsequent sections. It is important to note that these two paradigms form the backbone of philosophical science of qualitative research.